SEPSIS EARLY MANAGEMENT WORKFLOW ALGORITHM

KAISER PERMANENTE

SBC - Kaiser Permanente Fontana Medical Center / Kaiser Permanente Ontario Medical Center

IDENTIFICATION: 1-3 HOURS SEVERE SEPSIS PRESENTATION

Age ≥18 y.o.
Suspected infection

2+ SIRS Criteria:
- T<96.8 F or >100.9 F
- HR>90
- RR>20
- WBC <4K or >12K or >10% bands

CMS 1+ Organ Dysfunction Criteria:
- *Two SBP<90 / MAP<65 or SBP ↓>40 mmHg* w/in 3 hrs. of each other
- Acute Respiratory Failure – New need for invasive / non-invasive ventilation (BiPAP or CPAP)
- Cr >2 or U.O. <0.5 ml/kg/hr x 2 hrs.
- Total Bilirubin >2 mg/dL
- Platelet <100K
- INR >1.5 or aPTT >60 sec
- Lactate >2 mmol/L

qSOFA Criteria:
- Altered mental status (GCS<15)

SSP1: **Severe Sepsis Present** (Suspected inf., 2+ SIRS, & 1+ Organ Dysfunction) w/in 6 hrs. of each other?

No → Sepsis wo organ dysfunction (Suspected inf. 2+SIRS) → Standard Sepsis Therapy

Yes → Draw *INITIAL LACTATE* ≤1 hr. of SSP1 → Collect *BLOOD CULTURES* prior to first antibiotic administration ≤3 hrs. of SSP1 → Administer *BROAD SPECTRUM ANTIBIOTIC** ≤3 hrs. of SSP1

Initial lactate level 2.1-3.9 mmol/L → *REPEAT LACTATE* ≤3 hrs. of first sample collection time (≤6 hrs. of SSP1)

SSP2: Initial lactate level ≥4 mmol/L → *REPEAT LACTATE* ≤3 hrs. of first sample collection time (≤6 hrs. of SSP1)

SEPTIC SHOCK IF CLINICALLY APPROPRIATE, DIAGNOSE

SEVERE SEPSIS BUNDLE

Within 1-3 hrs. of presentation:
- Measure initial serum lactate w/in 1 hr. of SIRS / Sepsis identification
- Blood cultures prior to ABX
- Administer broad spectrum IV ABX

Within 3-6 hrs. of presentation:
- Repeat serum lactate if initial lactate >2 mmol/L

KP HEALTHCONNECT SEPSIS ORDER SETS
- ED ORDER SET
  - ADULT FEVER SEPSIS ED SCAL NATL [2601]
- IP ORDER SET
  - SEPSIS ICU IP SCAL NATL [3226]

INITIAL HYPOTENSION

TREATMENT / MGT.: 1-3 HOURS SEPTIC SHOCK PRESENTATION

Initial Hypotension Two SBP <90 / MAP<65 or SBP ↓>40 mmHg w/in 3 hrs. of each other?

No → Standard Sepsis Therapy

Yes → Administer *CRYSTALLOID IV FLUID BOLUS (30 ML/KG)* ≤3 hrs. of initial hypotension / SSP2 → Obtain *TWO COMPLETE SETS* of *VITAL SIGNS* (T, P, RR, & BP) 15 mins. apart within *1 HR. of IV FLUID COMPLETION TIME*

SSP2: **Persistent Hypotension** (Two consecutive SBP<90 / MAP<65 or SBP ↓ >40 mmHg) ≤1 hr. of IV fluid completion time?

Yes → *SSP2:* **Septic Shock Present**
- Severe Sepsis & Persistent Hypotension

OR
- Severe Sepsis & Initial Lactate ≥ 4 mmol/L

→ Administer *IV VASOPRESSOR* ≤6 hrs. of SSP2

No → Continue to EXCEPTIONS / EXCLUSIONS FOR SEPTIC SHOCK BUNDLE

SEPTIC SHOCK BUNDLE

Within 1-3 hrs. of presentation: Specify IVF bolus rate / duration
- 30 ml/kg crystalloid IVF resuscitation for initial hypotension or initial lactate ≥4 mmol/L. For BMI>30, use IBW: *{.sepsisIBWfluidvariation}*

Within 2-6 hrs. of presentation:
- Vasopressor administration for persistent hypotension
- Repeat volume status & tissue perfusion assessment

Focused reassessment exam s/p IVF bolus completion *{.EDHELP}*

EXCEPTIONS / EXCLUSIONS FOR SEPTIC SHOCK BUNDLE:

If *initial hypotension* or *elevated lactate* is normal for patient or related to a chronic condition or medication, please *specify* the *reason* and include "*due to non-infectious etiology*" or "*not sepsis-related*"

REASSESSMENT: 2-6 HOURS SEPTIC SHOCK PRESENTATION

Perform / Document:
- *TISSURE PERFUSION ASSESSMENT* (Complete Focused Reassessment Exam) ≤6 hrs. of SSP2 *{.EDHELP}*

OR
- *VOLUME STATUS TEST* ≤6 hrs. of SSP2

→ Continue critical sepsis care

Measure / Perform *ONE* **Volume Status Test**:
- Central Venous Pressure (CVP)
- Central Venous Oxygen Saturation (SvO2 or ScvO2)
- Echocardiogram or Cardiac Ultrasound (NICOM Cheetah)
- Fluid Challenge
- Passive Leg Raise

EXCEPTIONS / EXCLUSIONS TO SEPSIS BUNDLE TIMELINES:
- Directive for Comfort Care / Palliative Care w/in 6 hrs. of SSP1 / SSP2
- Patient refusal of any portion of the sepsis bundle
- EPRP / Acute Care Hospital Transfer
- Discharge within 6 hrs. of SSP1 / SSP2
- Documentation of viral, fungal, or parasitic infections

***SBC BROAD SPECTRUM ABX MONOTHERAPY**

TRADE NAME	GENERIC NAME
Rocephin	Ceftriaxone
Unasyn	Ampicillin / Sulbactam
Zosyn	Piperacillin / Tazobactam
Fortaz	Ceftazidime
Maxipime	Cefepime
Claforan	Cefotaxime (not in pyxis)

Add'l Abx for gram + / anaerobic coverage may be added if clinically indicated to these single agents

GET AHEAD OF SEPSIS
KNOW THE RISKS. SPOT THE SIGNS. ACT FAST.

SBC SEPSIS COORDINATORS:
FMC - Jennifer Platt, RN 909.302.6891
OMC - Judy Curran, RN 909.724.3559

LEGEND:
SSP1 – Severe Sepsis Presentation Time
SSP2 – Septic Shock Presentation Time

7.17.2020